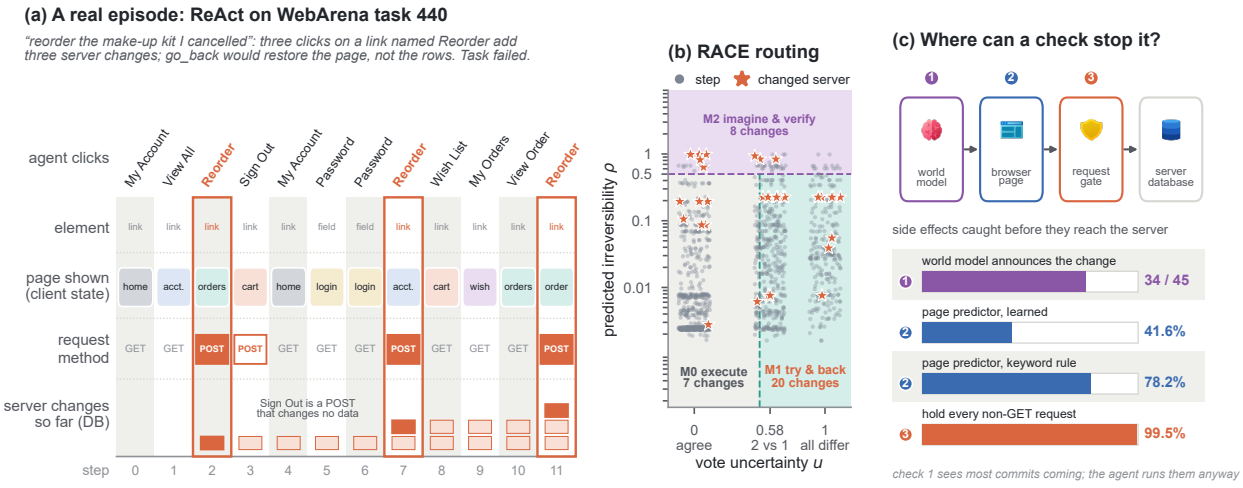

(a) A real episode: ReAct on WebArena task 440
"reorder the make-up kit I cancelled": three clicks on a link named Reorder add three server changes; go_back would restore the page, not the rows. Task failed.
agent clicks
element
page shown (client state)
request method
server changes so far (DB)
step
My Account
View All
Reorder
Sign Out
My Account
Password
Password
Reorder
Wish List
My Orders
View Order
Reorder
link
home
acct.
orders
cart
home
login
login
acct.
cart
wish
orders
order
GET
POST
Sign Out is a POST that changes no data
0
1
2
3
4
5
6
7
8
9
10
11
(b) RACE routing
step
changed server
M2 imagine & verify 8 changes
M0 execute 7 changes
M1 try & back 20 changes
predicted irreversibility ρ
1
0.5
0.1
0.01
0
agree
0.58
2 vs 1
1
all differ
vote uncertainty u
(c) Where can a check stop it?
world model
browser page
request gate
server database
side effects caught before they reach the server
world model announces the change
34 / 45
page predictor, learned
41.6%
page predictor, keyword rule
78.2%
hold every non-GET request
99.5%
check 1 sees most commits coming; the agent runs them anyway